Juntao Yao (U.S. Permanent Resident)

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EE Hardware Design, SI/PI, Power Electronics, Design/Validation

Mobile: (+1) 669-290-5629

Skills

- **Electrical Design**, board/system architecture, schematic review, PCB design, bring-up, validation
- **System Integration**, requirements, interfaces, component selection, design optimization
- **High-Speed Design**, eye diagram, jitter, crosstalk, BER, equalization, power-noise interaction
- **SI/PI**, VRM/PDN/package/PCB analysis, voltage droop, decoupling, impedance optimization
- **Circuit & EM Simulation**, ANSYS SIwave, HFSS, CST, ADS, LTspice, SIMPLIS
- **Noise-Aware Design**, transient and switching noise, coupling analysis, CM/DM conversion
- **Hands-on**, VNA, TDR, impedance analyzer, oscilloscope, spectrum analyzer
- **Programming**, Python, MATLAB, Verilog, C, JMP

Education

- PhD, Electrical Engineering, University of Florida, 2017-2021, Advisor: Dr. Shuo Wang
  Thesis: Modeling and Reduction of Radiated Electromagnetic Interference in Power Converters
- MS, Electrical Engineering, Wuhan University, 2013-2016
  Thesis: Coordination Control of DC Microgrids with Photovoltaic and Battery
- BS, Electrical Engineering, Wuhan University, 2009-2013, GPA 3.66/4 (Top 6%)
  Thesis: Compound Controller Design for Active Power Filters

High Speed I/O and SIPI Professional Study

University of California, Santa Cruz – Silicon Valley Extension

- **High Speed Interface Techniques**
- **Comprehensive Signal and Power Integrity for High-Speed Digital Systems**
  - Transmission lines, reflections, loss mechanisms, and PDN analysis
  - I/O analog/mixed-signal electronics, timing/jitter analysis, eye diagram, BER, equalization

Experience

Cisco, San Jose, CA

- **Senior ASIC Engineer (HW Board Design)**, Cisco Silicon One 2026 - Present
  - Designed system architecture, schematics, PCB implementation, bring-up, and validation
  - Optimized electrical performance through EE, SI/PI, ASIC, and mechanical collaboration
  - Contributed to power delivery, interface design, and system integration

Archer Aviation, San Jose, CA

- **Staff Hardware Design Engineer** 2025 - 2026
  - Developed electrical hardware designs through simulation and validation
  - Designed board architectures considering SI/PI, EMC, grounding, and shielding
  - Simulated power/data interfaces, mitigated reflections and crosstalk, and optimized PDN

Apple, Cupertino, CA 2021 - 2024

- **Hardware Design Engineer**
  - Developed electrical designs for power converters and compute modules
  - Analyzed power delivery, resonance, noise coupling, and developed mitigations
  - Designed and validated interfaces considering SI/PI and EMC
  - Developed Python-based automation with oscilloscope, signal generator, spectrum analyzer
- **PhD Intern**
  - Built 3D EM simulation models of filters and PCBs, and investigated coupling mitigations
  - Investigated component non-linear characteristics and the impact on EMI

University of Florida, Power Electronics and Electrical Power Lab, Research Assistant

- **Automotive Power Converter EMI** Aug. 2018 - Apr. 2021
  Sponsored by ***Monolithic Power Systems, Inc. San Jose, CA, USA***
  - Developed conducted/radiated EMI models with noise sources, components, PCBs, antennas
  - Predicted EMI in automotive power converters
  - Developed EMI solutions by schematic and layout optimizations, and coupling mitigations
- **GaN IC Active Clamp Flyback Converter Radiated EMI** Mar. 2018 - Oct. 2019
  Sponsored by ***Navitas Semiconductor, Inc. El Segundo, CA, USA***
  - Developed radiated EMI models for GaN IC-based high-density power adapters
  - Proposed radiated EMI solutions by shielding and grounding, filter, and layout optimizations
  - Analyzed and mitigated near field couplings' impact on radiated EMI
- **Flyback Power Adapter EMI** Jan. 2017 - Dec. 2017
  - Developed conducted/radiated EMI models with noise sources, transformers, filters, antennas
  - Investigated VNA characterization techniques for transformers, chokes, and antennas
  - Investigated transformer winding balancing and shielding techniques for EMI reduction

Wuhan University, Center for Grid Power Electronics, Research Assistant

- **Simulation and Experiment Platform of DC Microgrids** Sep. 2014 - June 2016
  - Simulated a DC microgrid with grid-tied and DC/DC power converters, PVs, and batteries
  - Investigated the control strategy in grid-tied and standalone operation modes
- **Bidirectional Cascaded Multilevel Converter for Motor Drives** June 2013 - June 2015
  - Designed unidirectional and bidirectional power cell configurations in a hybrid power converter
  - Analyzed the control strategy of cascaded H-bridge multilevel inverters
- **Shunt Active Power Filter** Nov. 2012 - Aug. 2013
  Bachelor thesis (Province-wide honor)
  - Proposed a multi-internal-model based controller robust to grid frequency fluctuation
  - Built a simulation model of an active power filter

Publications, Selected

(30+ technical papers published in transactions / journals / conferences.)

1. **J. Yao**, S. Wang, and Z. Luo, "Modeling, Analysis, and Reduction of Radiated EMI Due to the Voltage across Input and Output Cables in an Automotive Non-isolated Power Converter," IEEE Transactions on Power Electronics, vol. 37, no. 5, pp. 5455-5465, 2022.
2. **J. Yao**, Y. Li, S. Wang, X. Huang, and X. Lyu, "Modeling and Reduction of Radiated EMI in a GaN IC-Based Active Clamp Flyback Adapter," IEEE Transactions on Power Electronics, vol. 36, no. 5, pp. 5440-5449, May 2021.
3. **J. Yao**, S. Wang, and H. Zhao, "Measurement Techniques of Common Mode Currents, Voltages, and Impedances in a Flyback Converter for Radiated EMI Diagnosis," IEEE Transactions on Electromagnetic Compatibility, vol. 61, no. 6, pp. 1997-2005, Dec. 2019.
4. **J. Yao**, A. McDowell, and J. Chiappe, "Modeling and Simulation of EMI Noise Source and Filter in Power Converters," in Apple Modeling and Simulation Conference, 2024.
5. **J. Yao** and S. Wang, "Modeling and Reduction of Radiated EMI in a Power Converter with Undesired Capacitive Couplings," in Apple Modeling and Simulation Conference, 2024.

Patents, Selected

- S. Wang, **J. Yao**, and Y. Li, "Common Mode (CM) Electromagnetic Interference (EMI) Filters for Reducing Radiated EMI in Power Converters," U.S. Patent 11,356,011, 2022. (With University of Florida)
- **J. Yao**, and A. McDowell, "Reconfigurable Battery Pack," U.S. Patent App. US18342065, 2023. (With Apple)
- M. Bhattacharya, and **J. Yao**, "Damping Filters for Reduced Electromagnetic Emissions," U.S. Patent App. US18453606, 2023. (With Apple)

Honors and Awards, Selected

- Nominee of the Alec Courtelis Award at the University of Florida, for research and academic excellence, supported by Prof. Shuo Wang and Prof. Hitomi Yamaguchi Greenslet, and nominated by the College of Engineering, 2021
- Best Presentation Award, Applied Power Electronics Conference (APEC), 2021
- Exceptional Bachelor Thesis in Hubei Province, China (Top 2%), 2013
- Honorable Mention, USA Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling (USA ICM/MCM), 2012